

Training: tcn_lite (selected family)

The full causal TCN, 24 filters, and the training of the selected model tcn_lite_ft.

The deployed model tcn_lite_ft_int8 was trained in two stages: tcn_lite from scratch in round 1 (48 full passes over the 20.8M training windows, 9.5 hours, ending at the 1e-5 learning-rate floor), then a fine-tune of those weights at lr 3e-4 (7 epochs of 2M windows; the best was the first, so the weights barely moved). Finally a best-of-K int8 export.

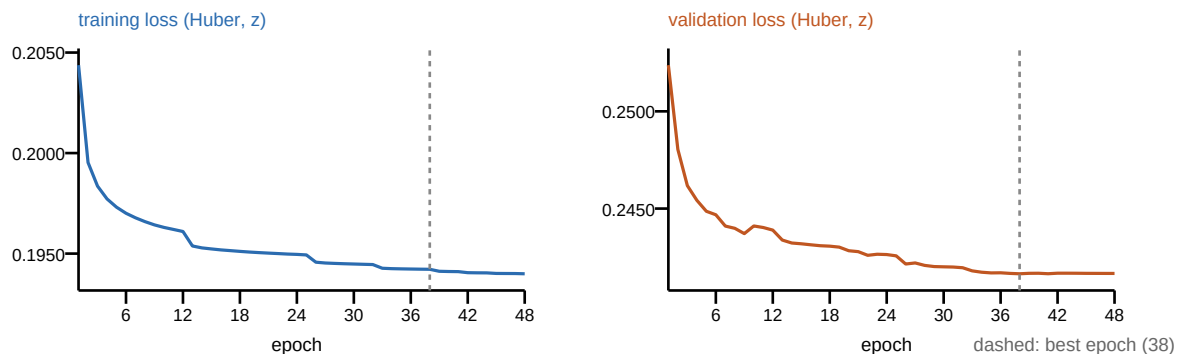
How this family is trained

- Inputs: the (30, 2) z-scored RR/dRR sequence + the 9 'seq' features (rmssd, rs, ccm, guzik, rls_pred, rls_err, bp, bp_prev, run_len); residual target; the shared Keras recipe; BatchNorm folded at export.
- tcn_lite_ft was planned as the control of tcn_lite_dad: the same fine-tune schedule without closed-loop data, to separate the effect of DAD from that of restarting the learning rate.

Runs

tcn_lite - round 1 base

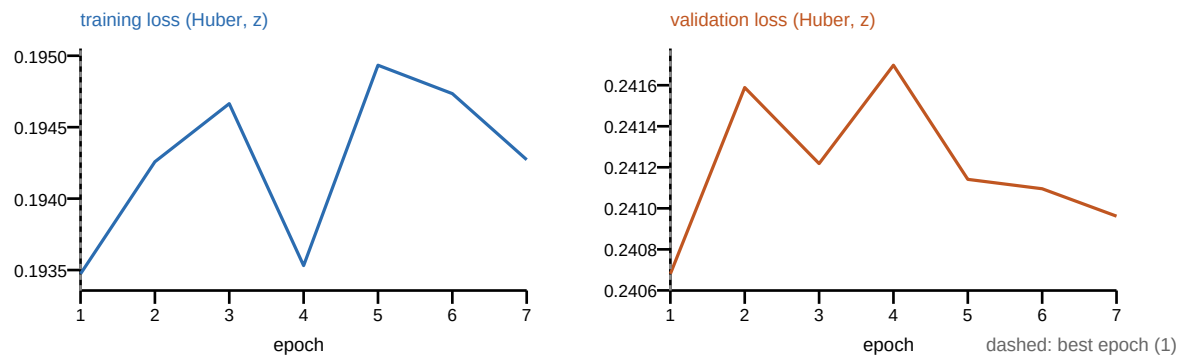
item	value
command	python src/train.py --models tcn_lite --tag tcn_lite
start	from scratch (random initialisation, seed 211)
inputs	30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	20,755,456 windows = one full pass over the 20,755,925 training windows (round-1 train.py)
budget	48 of at most 100 epochs (early stopping, patience 10); 996,261,888 windows drawn = 48.0 passes over the training set
learning rate	0.001 at the start, halved 7 times on validation plateaus, 1e-05 at the end
best epoch	38 (validation loss 0.2524 after epoch 1, 0.2416 at best); its weights are the saved model
time	9.49 h on one GPU
int8	re-exported 2026-10-08 with best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.26% on the held-out training windows (plain 20k min/max: +0.54%); full validation: int8 33.01 vs float 32.64 ms (+1.13%)
validation RMSE (float)	32.64 ms on all 1,421,225 validation windows (-12.1% vs rls_ar 37.14)



- Re-exported with 20,000 windows and then best-of-K on 2026-10-08.

tcn_lite_ft - **SELECTED: fine-tune, control for DAD**

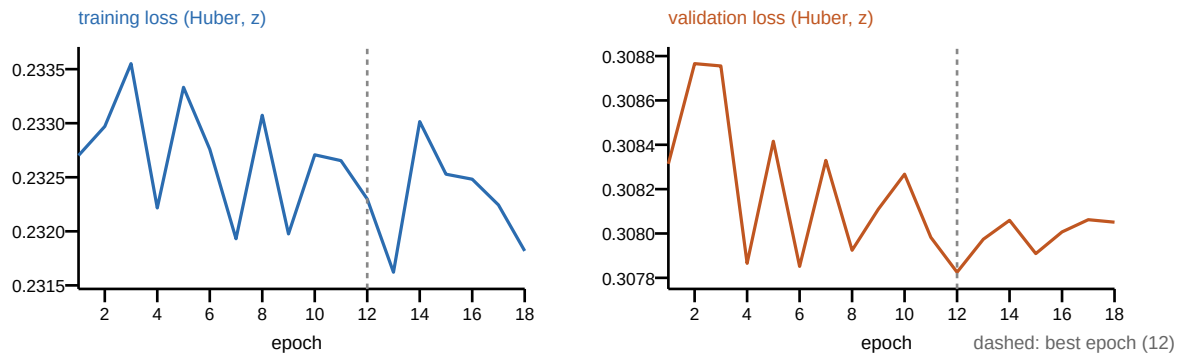
item	value
command	python src/train.py --models tcn_lite --init_from runs/tcn_lite --lr 0.0003 --epochs 30 --patience 6 --tag tcn_lite_ft
start	fine-tune of tcn_lite: its weights, features, residual flag and normalisation; a fresh AdamW optimiser
inputs	30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	7 of at most 30 epochs (early stopping, patience 6); 13,999,104 windows drawn = 0.7 passes over the training set
learning rate	0.0003 at the start, halved 2 times on validation plateaus, 7.5e-05 at the end
best epoch	1 (validation loss 0.2407 after epoch 1, 0.2407 at best); its weights are the saved model
time	0.11 h on one GPU, 14.2 ms per step
int8	re-exported 2026-10-08 with best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.31% on the held-out training windows (plain 20k min/max: +0.51%); full validation: int8 32.98 vs float 32.62 ms (+1.08%)
validation RMSE (float)	32.62 ms on all 1,421,225 validation windows (-12.2% vs rls_ar 37.14); parent 32.64 ms



- The first epoch at lr 3e-4 already gave the best validation loss (0.2407 vs 0.2417 for the parent); the next 6 did not improve it and early stopping restored epoch 1. The gain over tcn_lite is tiny (0.02 ms open loop) but consistent in the int8 closed loop (pooled, 14/16 subjects).
- int8: draw n2000_s2 (2,000 windows), md5 6219d6ae...; this file is the one evaluated on the test split.

tcn_lite_dad - **round 3, fine-tune on closed-loop data**

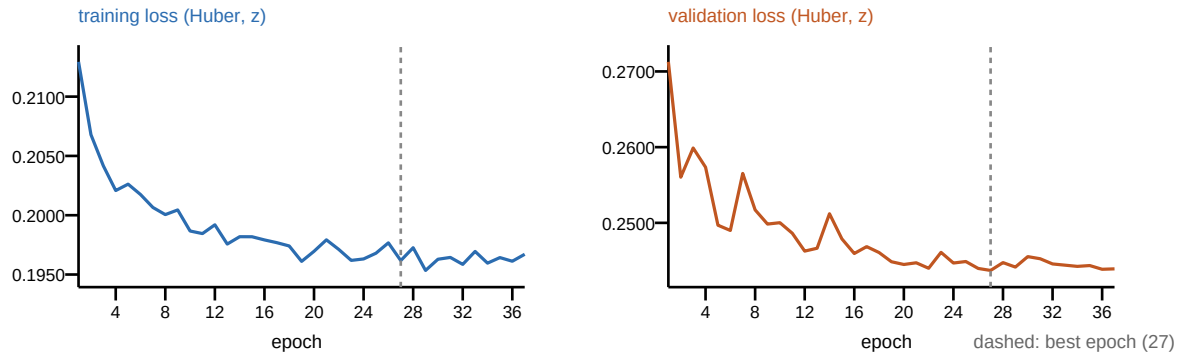
item	value
command	python src/train.py --models tcn_lite --init_from runs/tcn_lite --dad_filler runs/tcn_lite --dad_share 0.5 --lr 0.0003 --epochs 30 --patience 6 --tag tcn_lite_dad
start	fine-tune of tcn_lite: its weights, features, residual flag and normalisation; a fresh AdamW optimiser
inputs	30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	18 of at most 30 epochs (early stopping, patience 6); 35,997,696 windows drawn = 1.7 passes over the training set
learning rate	0.0003 at the start, halved 5 times on validation plateaus, 1e-05 at the end
best epoch	12 (validation loss 0.3083 after epoch 1, 0.3078 at best); its weights are the saved model
time	0.43 h on one GPU, 21.5 ms per step
closed-loop data	tcn_lite filled 4,220 training segments of 5000 beats (20,511,092 windows, 10%-80% dropped per segment, seed 0) in 297 s on CPU; each training window is closed-loop with probability 0.5 (realised 0.503); early stopping on a 50/50 mix of ordinary and closed-loop validation windows
int8	re-exported 2026-10-08 with best-of-K calibration: chose draw n2000_s1 (2000 windows), int8-vs-float gap +0.29% on the held-out training windows (plain 20k min/max: +0.57%); full validation: int8 33.37 vs float 33.09 ms (+0.84%)
validation RMSE (float)	33.09 ms on all 1,421,225 validation windows (-10.9% vs rls_ar 37.14); parent 32.64 ms; closed-loop validation windows 42.24 ms (parent 42.29, rls_ar 44.26)



- A first attempt (2026-10-08 01:07) drew only the first 429k windows of each half because of the tf.data index bug: training loss 0.083 and validation 56.76 ms, an overfit to the first subjects. It was deleted and this run re-queued after the fix.

tcn_lite_rsa - round 4, + gated RSA features, from scratch

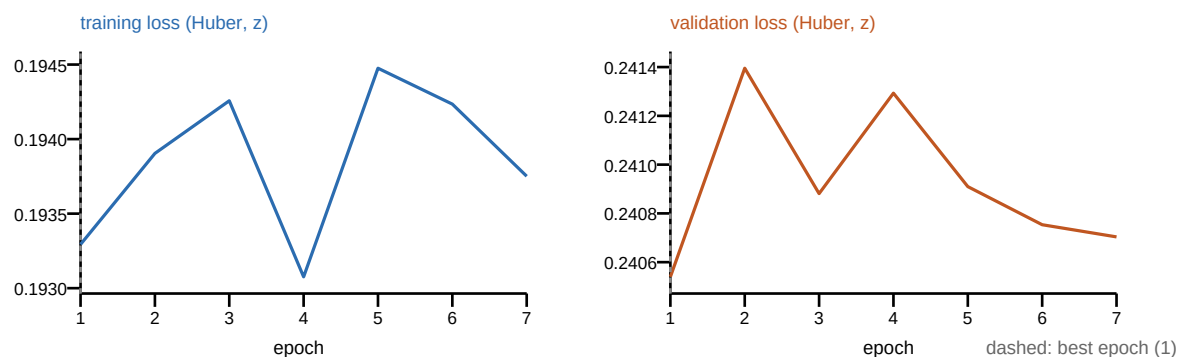
item	value
command	python src/train.py --models tcn_lite --features seq_rsa --tag tcn_lite_rsa
start	from scratch (random initialisation, seed 211)
inputs	30-beat RR sequence (z-RR, z-dRR) + 13 tabular features (set seq_rsa)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	37 of at most 100 epochs (early stopping, patience 10); 73,995,264 windows drawn = 3.6 passes over the training set
learning rate	0.001 at the start, halved 6 times on validation plateaus, 1.6e-05 at the end
best epoch	27 (validation loss 0.2713 after epoch 1, 0.2437 at best); its weights are the saved model
time	0.63 h on one GPU, 15.6 ms per step
int8	best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.16% on the held-out training windows (plain 20k min/max: +0.45%); full validation: int8 33.29 vs float 33.06 ms (+0.71%)
validation RMSE (float)	33.06 ms on all 1,421,225 validation windows (-11.0% vs rls_ar 37.14)



- About 3.6 passes vs 48 for its parent: the +0.42 ms is confounded by the smaller training budget.

tcn_lite_rsa_ws - round 4, + gated RSA features, warm start

item	value
command	python src/train.py --models tcn_lite --features seq_rsa --init_from runs/tcn_lite --lr 0.0003 --epochs 30 --patience 6 --tag tcn_lite_rsa_ws
start	warm start from tcn_lite: every weight copied, the 4 new inputs (rsa_f, rsa_amp, rsa_q, rsa_drr) start at zero weight, so at step 0 it computes exactly the parent's function (checked: max difference 1.9e-06 z on real validation windows)
inputs	30-beat RR sequence (z-RR, z-dRR) + 13 tabular features (set seq_rsa)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	7 of at most 30 epochs (early stopping, patience 6); 13,999,104 windows drawn = 0.7 passes over the training set
learning rate	0.0003 at the start, halved 2 times on validation plateaus, 7.5e-05 at the end
best epoch	1 (validation loss 0.2405 after epoch 1, 0.2405 at best); its weights are the saved model
time	0.12 h on one GPU, 15.1 ms per step
int8	best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.25% on the held-out training windows (plain 20k min/max: +0.36%); full validation: int8 33.00 vs float 32.62 ms (+1.16%)
validation RMSE (float)	32.62 ms on all 1,421,225 validation windows (-12.2% vs rls_ar 37.14); parent 32.64 ms



- Same schedule as tcn_lite_ft, so the two differ only by the 4 new inputs: no gain at 10-50% dropped and a small loss at 80%.

What the training showed

- tcn_lite reached the learning-rate floor with a flat validation loss: more epochs would not help, and the fine-tune confirms it (best epoch = first).
- Neither closed-loop data (DAD) nor RSA inputs gave a significant int8 closed-loop gain over tcn_lite_ft, which was selected and passed the single test evaluation.

Abbreviations and acronyms used in this document

term	meaning
bp	feature: RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1
bp_prev	feature: previous band-pass output (gives the phase with bp)
ccm	complex correlation measure: non-linearity index of the Poincare plot of consecutive RR intervals
DAD	data as demonstrator: training on inputs that contain the model's own closed-loop fills, with the true target (a remedy for exposure bias)
_dad	run-name suffix: fine-tuned with DAD (closed-loop-filled) training data
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
_ft	run-name suffix: fine-tuned (training continued from the parent's weights)
guzik	feature: Guzik asymmetry: sum of positive dRR / sum of dRR
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
lr	learning rate (step size of the optimiser)
ms	milliseconds
rls_err	feature: last RLS innovation: true dRR minus the previous forecast (ms)
rls_pred	feature: RLS AR(8) forecast of the next dRR (ms), clamped to +/-300
rmssd	root mean square of successive differences of the RR intervals in the window
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
rs	rescaled range: (max - min of the cumulative deviations from the mean) / standard deviation
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
run_len	feature: signed length of the current run of same-sign dRR (+ = decelerating)
seq	feature-set / model-kind name: sequence model (takes the raw RR window plus tabular features)
TCN	temporal convolutional network (causal, dilated 1-D convolutions)
tcn	temporal convolutional network (run-name part)
z	z-score: (value - mean) / standard deviation, with statistics from the training subjects